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RESEARCH

# Single-session extraoral photobiomodulation reduces early postoperative pain but not edema or trismus after impacted mandibular third molar extraction: a randomized controlled trial

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**Abstract**

Postoperative pain, edema, and trismus are frequent complications following impacted mandibular third molar surgery. While photobiomodulation (PBM) is a proposed adjunctive therapy, clinical evidence remains controversial. This study evaluated the effects of single-session extraoral dual-wavelength PBM (650 + 904 nm) applied immediately after extraction on postoperative pain, edema, trismus, and quality of life (QoL). This prospective RCT enrolled 70 patients (mean age: 22.04 ± 2.61 years) with asymptomatic mesioangular impacted mandibular third molars (Pell–Gregory Class IB). Participants were randomized to a PBM (n = 35) or control (n = 35) group. Following standardized surgical extraction by a single surgeon, the PBM group received a 10-minute extraoral application (650 + 904 nm, GaAlAs laser). The primary outcome was postoperative pain, assessed by VAS on postoperative days 2 and 7. Secondary outcomes included QoL (VAS, days 2 and 7), edema (sum of three facial distances), and trismus (maximum interincisal distance), the latter two evaluated at baseline and on postoperative days 2 and 7. On postoperative day 2, pain VAS scores were significantly lower in the PBM group (20.34 ± 24.38 vs. 41.43 ± 25.15; $p < 0.001$) and QoL scores significantly higher (69.83 ± 25.87 vs. 50.51 ± 23.24; $p = 0.002$) compared to controls. By day 7, a statistically significant but clinically negligible between-group difference persisted in pain (Mann-Whitney $U = 776.5$, $p = 0.040$; mean difference 2.6 mm), while QoL no longer differed significantly between groups ($p = 0.154$). No significant between-group differences were observed for edema or trismus at any time point ($p > 0.05$). Single-session extraoral dual-wavelength PBM significantly reduces early postoperative pain and improves QoL on the second postoperative day following impacted third molar extraction, with no significant effect on edema or trismus. The clinically meaningful pain and QoL benefits resolved by day 7, and sham-controlled multi-session trials are warranted.

**Keywords** Photobiomodulation · Impacted third molar · Postoperative pain · Edema · Trismus · Quality of life · Randomized controlled trial

## Introduction

Surgical removal of impacted mandibular third molars is among the most frequently performed procedures in oral and maxillofacial surgery. Postoperative pain, edema, and trismus resulting from surgical trauma can adversely affect patients' daily activities and quality of life, leading to various social and psychological problems [1]. The management of these complications continues to be a matter of clinical significance.

Numerous pharmacological and non-pharmacological approaches have been investigated to reduce postoperative morbidity. These include systemic and local corticosteroids,

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non-steroidal anti-inflammatory drugs, cryotherapy, platelet-rich fibrin-based materials, and photobiomodulation (PBM) [2, 3]. PBM is thought to exert its effects through mitochondrial photoactivation by enhancing ATP synthesis, modulating reactive oxygen species, and reducing proinflammatory mediators such as prostaglandin $E_2$, interleukin-1, and tumor necrosis factor-alpha. It has been suggested that PBM may alleviate these complications in the early postoperative period following third molar surgery by suppressing the inflammatory cascade at the mitochondrial level, promoting lymphatic drainage, and reducing peripheral nerve conduction velocity [4–6].

Nevertheless, the clinical evidence regarding the efficacy of PBM following third molar surgery remains inconsistent. While some systematic reviews and meta-analyses have reported statistically significant benefits in terms of pain and edema [7], others have failed to demonstrate a significant advantage over control groups, particularly with respect to trismus [8]. Individual studies have also yielded conflicting results: a split-mouth study reported significantly lower perceived edema and pain scores in the early postoperative period [9], whereas a multicenter triple-blind study found no statistically significant differences in any parameter [10]. This heterogeneity in the literature is largely attributed to variations in laser wavelength, energy dose, application site, timing, and number of sessions [11]. Notably, very few studies have evaluated the effects of a single-session extraoral PBM protocol applied immediately after surgery.

The aim of this study was to evaluate the effects of a single-session dual-wavelength extraoral PBM (650 nm + 904 nm) applied immediately after the surgical extraction of asymptomatic mesioangular impacted mandibular third molars on postoperative pain, edema, trismus, and quality of life, using a randomized controlled clinical trial design.

## Materials and methods

### Study design and ethical approval

This prospective, parallel-group, randomized controlled clinical trial was designed and reported in accordance with the Consolidated Standards of Reporting Trials (CONSORT) 2010 guidelines [12] (Fig. 1). The study was conducted at a university-affiliated tertiary care center (anonymized for peer review). The study protocol was approved by the Institutional Ethics Committee for Non-Drug and Non-Medical Device Research (anonymized for the review process) (Decision No: 2025/535, Date: January 30, 2025). Written informed consent was obtained from all participants prior to enrollment.

### Patient selection

Patients presenting with an asymptomatic impacted mandibular third molar in the mesioangular position according to Winter's classification and in Class IB position according to the Pell–Gregory classification on panoramic radiography were enrolled. Following the initial screening, simple randomization was performed to allocate participants into either the PBM group ($n = 35$) or the control group ($n = 35$) using a computer-generated random sequence. Allocation concealment was achieved using sequentially numbered, opaque, sealed envelopes. Through this process, a total of 70 patients (28 males, 42 females; age range: 17–25 years; mean age: 22.04 ± 2.61 years) were included. The PBM group comprised 13 males and 22 females, while the control group comprised 15 males and 20 females. To ensure scientific rigor, the clinical investigator performing the postoperative measurements was blinded to the group allocations.

### Inclusion and exclusion criteria

The following inclusion criteria were: (i) indication for extraction of an impacted mandibular third molar in the mesioangular position, (ii) absence of systemic disease (ASA I status), (iii) healthy status of the impacted tooth and surrounding tissues, (iv) adequate oral hygiene, and (v) absence of malignant or premalignant oral lesions.

Patients meeting any of the following criteria were excluded: (i) pregnancy or lactation, (ii) regular use of non-steroidal anti-inflammatory drugs or antibiotics within the preceding week, (iii) presence of a diagnosed psychiatric disorder, and (iv) active smoking.

### Surgical procedure

All surgical procedures were performed by the same surgeon —a specialist in oral and maxillofacial surgery with five years of clinical experience— following a standardized protocol. Local anesthesia was achieved by inferior alveolar and buccal nerve blocks using 4% articaine hydrochloride with 1:100,000 epinephrine (Ultracain DS Forte; Sanofi-Aventis, Istanbul, Türkiye). A crestal horizontal incision was made with a No. 15 scalpel blade, and a vertical releasing incision was placed at the mesiobuccal aspect of the adjacent second molar to raise a full-thickness mucoperiosteal flap.

Bone removal and tooth sectioning were carried out under copious irrigation using tungsten carbide burs; a round bur was used for ostectomy around the impacted tooth, and a fissure bur was used for tooth division. Sharp bony edges were smoothed, follicular remnants were curetted, and the extraction socket was irrigated thoroughly with sterile

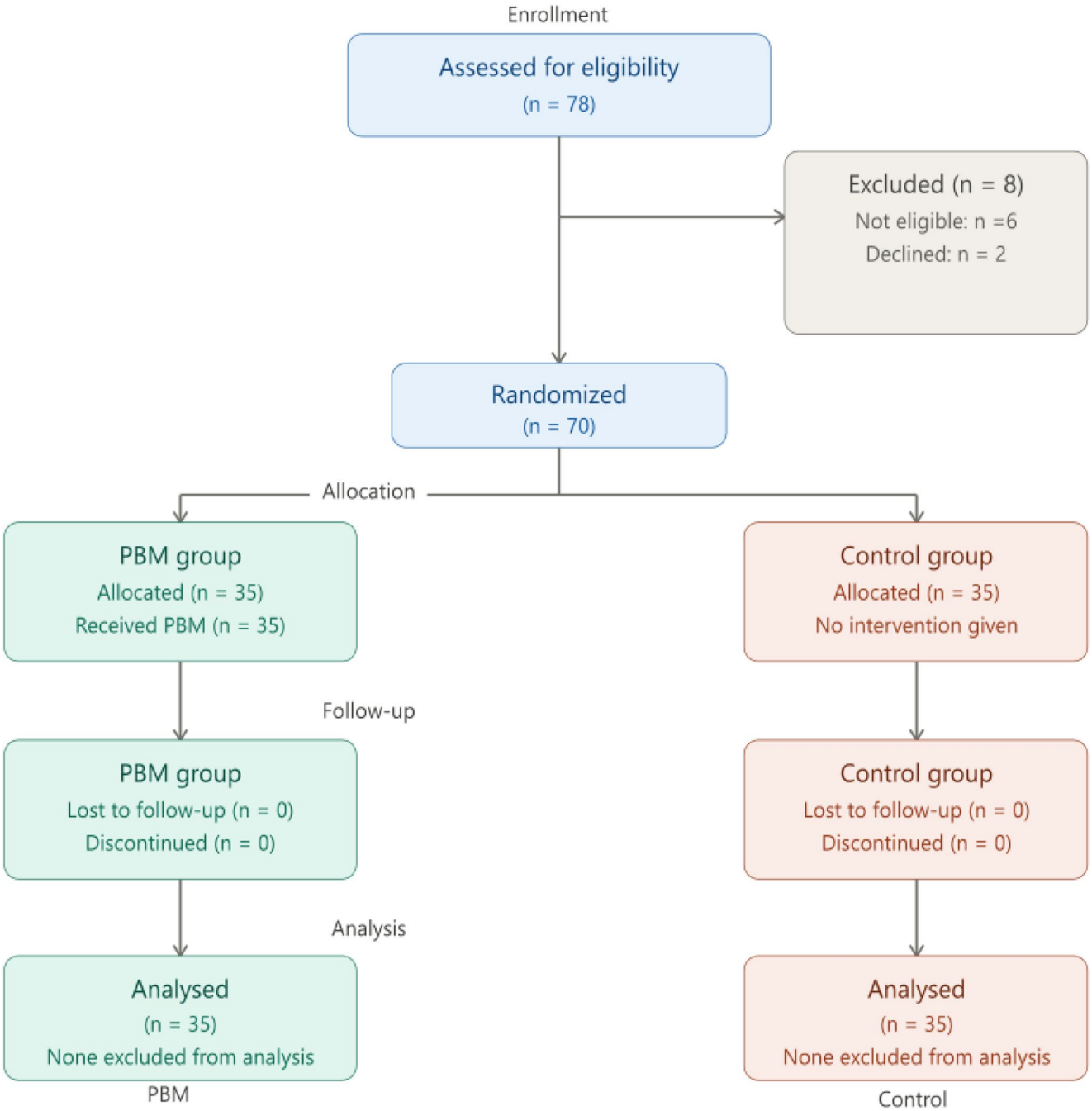



**Fig. 1** The CONSORT flow diagram illustrating participant screening, exclusion, randomization, and analysis

saline. Following hemostasis, the flap was repositioned and closed primarily with 3−0 silk sutures (Dogsan, Trabzon, Türkiye). Sutures were removed atraumatically on the seventh postoperative day. The operative time, measured from the first incision to the final suture, was recorded for each procedure to ensure there were no significant differences in surgical difficulty between the groups.

All patients received the same postoperative medication regimen: amoxicillin (875 mg) and clavulanic acid (125 mg) twice daily (Augmentin BID; GlaxoSmithKline, Istanbul, Türkiye), a benzydamine HCl and chlorhexidine gluconate mouth rinse three times daily (Kloroben; Drogsan, Ankara, Türkiye), and dexketoprofen 25 mg twice daily (Arveles; Ufsa, Istanbul, Türkiye). Patients were permitted to use paracetamol (500 mg) as a rescue analgesic if the prescribed regimen was insufficient, and all such intakes were recorded. Standardized postoperative care instructions were provided to all patients, and any

occurrence of postoperative complications, such as alveolar osteitis or infection, was monitored throughout the follow-up period.

## Photobiomodulation (PBM) protocol

Participants in the PBM group received a single session of extraoral photobiomodulation therapy (PBMT) immediately following the final suture. The treatment was administered using a dual-wavelength gallium aluminum arsenide (GaAlAs)-based laser system combined with light-emitting diodes (LEDs) (GRR Laser; GRR Laser Medical Ltd., Ankara, Türkiye). The device operated at wavelengths of 650 nm (7 LEDs) and 904 nm (8 LEDs). The extraoral probe, with an active emission area of 99 cm2 (110 mm × 90 mm), was positioned in stationary contact over the mandibular angle–ramus region on the operated side.

The application lasted for 10 min, delivering a total energy of 270 J (27 J/min). Based on the active probe area, the protocol provided a power density (irradiance) of 4.54 mW/cm2 and a total energy density (fluence) of 2.73 J/cm2 per session. According to the manufacturer's specifications, the system provides a tissue penetration depth of up to 50 mm (Fig. 2).

The probe was disinfected with 70% isopropyl alcohol between patients, and no additional intervention was provided to the control group. To ensure the consistency of energy delivery, the device's output was verified by the manufacturer prior to the study commencement.

## Outcome variables

Postoperative pain served as the primary outcome, on which the sample size calculation was based. Edema, trismus, and oral health-related quality of life (OHRQoL) were evaluated as secondary outcomes; it is acknowledged that would be required to ensure adequate statistical power for each secondary endpoint individually.

### Pain

Postoperative pain was assessed using a 100-mm visual analog scale (VAS; 0 = no pain, 100 = worst imaginable pain) at postoperative day 2 (T1) and day 7 (T2). To account for potential confounding factors, any intake of rescue analgesics (paracetamol 500 mg) was recorded by the patients in a diary and verified during follow-up visits to ensure the accuracy of the pain reports.

### Edema

Postoperative edema was quantified by measuring the distances between three pairs of anatomical reference points using a flexible tape measure: tragus to labial commissure (T–LC), gonion to lateral canthus (GO–C), and gonion to labial commissure (GO–LC) [5, 13, 14] (Fig. 3). To ensure objective evaluation and minimize measurement bias, all measurements were performed by a single blinded investigator (anonymized for review) who was unaware of the group




**Fig. 2** Extraoral application of the photobiomodulation device

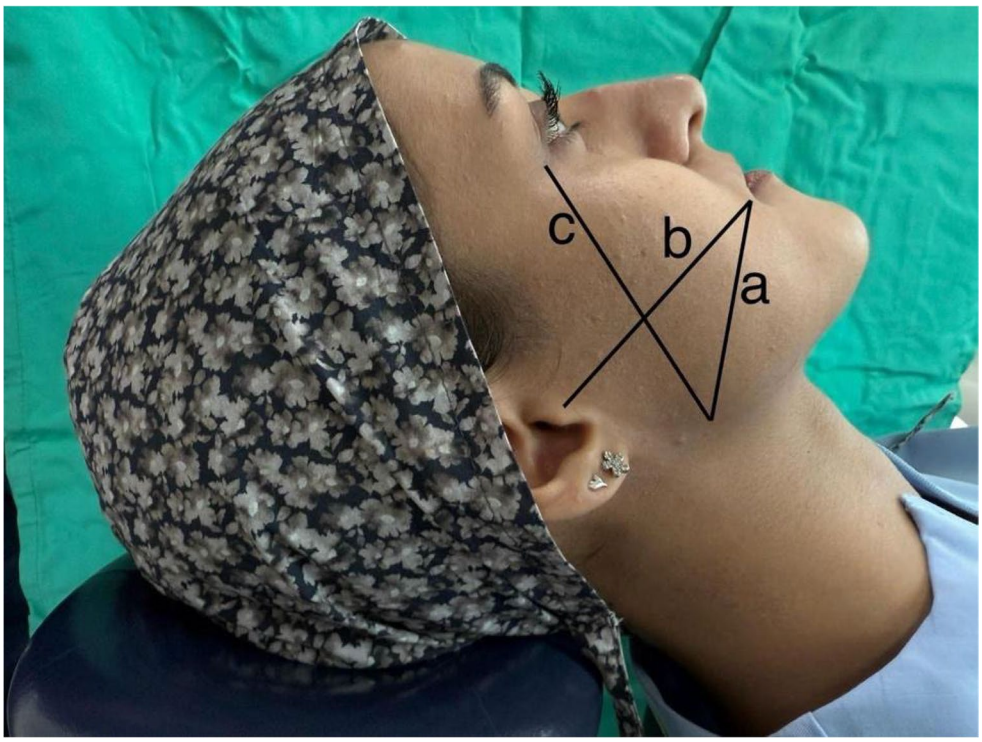



**Fig. 3** Anatomical landmarks used for facial distance measurements. (**A**) Gonion–labial commissure, (**B**) Tragus–labial commissure, (**C**) Gonion–lateral canthus

allocations. Measurements were recorded at three time points: preoperatively (T0), postoperative day 2 (T1), and postoperative day 7 (T2). Results are presented as the sum of the three facial distances in millimeters at each time point.

### Trismus

Trismus was quantified by measuring the maximum interincisal distance (MID) between the incisal edges of the maxillary and mandibular central incisors during maximum active mouth opening. Measurements were recorded at T0 (baseline), T1 (postoperative day 2), and T2 (postoperative day 7) using a standardized sliding caliper with millimeter precision. To ensure objective evaluation and minimize assessment bias, all measurements were performed by a single blinded investigator who was unaware of the group allocations. To ensure clinical consistency, three consecutive readings were taken at each time point, and the average value was used for statistical analysis.

### Quality of life

The impact of the surgical procedure and the subsequent recovery on the patients' daily lives was evaluated using a global Oral Health-Related Quality of Life (OHRQoL) scale adapted from previously published methodology [15]. This was assessed via a 100-mm visual analog scale (VAS), where 0 represented the "worst possible quality of life" and 100 represented "normal quality of life" (comparable to the preoperative state).

Patients were instructed to provide a global assessment of their overall well-being, including their ability to perform daily activities, social interactions, and functional tasks (such as eating and speaking) at T1 (postoperative day 2) and T2 (postoperative day 7). To ensure the independence of the scores, the OHRQoL assessment was completed by the patients separately from the pain VAS.

### Statistical analysis

Sample size calculation was performed prior to the study using G*Power software (version 3.1.9.4; Heinrich Heine University, Düsseldorf, Germany). Based on a pilot study, a minimum of 30 patients per group was required to achieve a power of 80% with a 5% significance level to detect a clinically significant difference in postoperative pain (primary outcome). To account for potential dropouts, 35 patients were enrolled per group.

Descriptive statistics were calculated for all variables and presented as mean ± standard deviation (SD) or frequencies. The normality of data distribution was assessed using the Shapiro–Wilk and Kolmogorov–Smirnov tests. Baseline demographic data (age and sex) were compared between groups using the independent samples t-test and chi-square test, respectively, to confirm successful randomization.

For the primary longitudinal analysis, a two-way mixed analysis of variance (ANOVA) was applied to each

**Table 1** Between-group comparison results at all time points

| Parameter | Time point | Control (Mean ± SD) | PBM (Mean ± SD) | *p* value |
|---|---|---|---|---|
| MMO (mm) | 0. Day | 43.87 ± 6.60 | 46.00 ± 6.91 | 0.195 |
| | 2. Day | 24.40 ± 8.20 | 26.09 ± 7.56 | 0.251 |
| | 7. Day | 34.89 ± 6.81 | 35.17 ± 6.74 | 0.860 |
| Pain VAS | 2. Day | 41.43 ± 25.15 | 20.34 ± 24.38 | < 0.001* |
| | 7. Day | 9.34 ± 11.16 | 6.74 ± 15.08 | 0.040* |
| Quality of Life VAS | 2. Day | 50.51 ± 23.24 | 69.83 ± 25.87 | 0.002* |
| | 7. Day | 87.89 ± 15.40 | 92.34 ± 12.93 | 0.154 |
| Edema (mm) | 0. Day | 286.54 ± 18.94 | 285.37 ± 17.88 | 0.791 |
| | 2. Day | 304.49 ± 18.77 | 301.34 ± 17.68 | 0.473 |
| | 7. Day | 292.71 ± 18.80 | 291.49 ± 16.67 | 0.897 |

* Statistically significant ($p < 0.05$)

continuous outcome variable, with group (PBM vs. control) as the between-subjects factor and time as the within-subjects factor. The group × time interaction was the primary inferential target. For outcome variables assessed at three time points (edema, maximum mouth opening), the sphericity assumption was evaluated using Mauchly's test; where sphericity was violated ($p < 0.05$), degrees of freedom were corrected using the Greenhouse–Geisser epsilon. Bonferroni correction was applied for pairwise comparisons following significant main effects or interactions. For outcome variables assessed at two time points only (pain VAS, QoL VAS), sphericity is automatically satisfied; within-group changes were examined with Wilcoxon signed-rank tests (non-normal distributions) or paired samples t-tests (normal distributions), and between-group differences at each time point were examined with Mann-Whitney U tests (non-normal distributions) or independent samples t-tests (normal distributions), as determined by Shapiro–Wilk testing.

All statistical analyses were conducted in Python using the following libraries: pingouin (v0.5), SciPy (v1.x), and NumPy. The level of statistical significance was set at $p < 0.05$.

## Results

### Descriptive statistics

A total of 70 patients aged between 17 and 25 years (mean age: 22.04 ± 2.61 years) were enrolled in the study. The control group consisted of 15 males and 20 females (mean age: 22.14 ± 2.40 years), while the PBM group consisted of 13 males and 22 females (mean age: 21.94 ± 2.83 years). No statistically significant differences were found between the groups with regard to age or sex distribution. The mean operative time was 24.8 ± 6.1 min in the control group and 25.7 ± 5.8 min in the PBM group; the difference between groups was not statistically significant (independent samples t-test; $p = 0.530$), indicating comparable surgical difficulty. Between-group comparison results for all outcome measures are presented in Table 1.

No patient reported using any rescue analgesic (paracetamol) or any medication other than the prescribed regimen during the seven-day follow-up period. No postoperative infection or alveolar osteitis (dry socket) was observed in any patient throughout the follow-up.

### Interaction effects

The two-way mixed ANOVA revealed a statistically significant group × time interaction for pain VAS [$F(1,68) = 9.644$, $p = 0.003$, partial $\eta^2 = 0.124$] and QoL VAS [$F(1,68) = 6.975$, $p = 0.010$, partial $\eta^2 = 0.093$]. No significant group × time interaction was detected for MMO [$F(2,136) = 0.737$, $p = 0.480$, partial $\eta^2 = 0.011$] or edema [$F(2,136) = 0.910$, $p = 0.405$, partial $\eta^2 = 0.013$]. Mauchly's sphericity test indicated violated sphericity for MMO ($W = 0.643$, $\chi^2(2) = 30.05$, $p < 0.001$, $\varepsilon(GG) = 0.737$) and edema ($W = 0.571$, $\chi^2(2) = 38.15$, $p < 0.001$, $\varepsilon(GG) = 0.700$); Greenhouse–Geisser correction was applied to the time main effect accordingly. Pain VAS and QoL VAS were each measured at two time points, satisfying sphericity automatically ($\varepsilon = 1.000$). Within-group and between-group pairwise comparisons are detailed below.

### Within-group comparisons

#### Control group

Maximum mouth opening (MMO) values in the control group revealed statistically significant differences among the time points (Bonferroni post-hoc; $p < 0.001$ for all pairwise comparisons: day 0 vs. day 2, day 0 vs. day 7, and day 2 vs. day 7). These findings indicate a marked reduction in MMO during the early postoperative period, followed by partial recovery by day 7.

Pain VAS scores decreased significantly from day 2 to day 7 (Wilcoxon signed-rank test; $p<0.001$). Quality of life VAS scores evaluated at the same time points showed a significant improvement on day 7 (Wilcoxon signed-rank test; $p<0.001$).

Edema measurements yielded significant differences across all pairwise comparisons (day 0 vs. day 2: $p<0.001$; day 2 vs. day 7: $p<0.001$; day 0 vs. day 7: $p<0.001$), indicating that postoperative edema peaked on day 2 and subsequently began to resolve.

### PBM group

Bonferroni post-hoc analysis for MMO values in the PBM group demonstrated statistically significant differences between all time points: day 0 vs. day 2 ($p<0.001$), day 0 vs. day 7 ($p<0.001$), and day 2 vs. day 7 ($p<0.001$).

Pain VAS scores decreased significantly from day 2 to day 7 (Wilcoxon signed-rank test; $p<0.001$). Quality of life VAS scores likewise improved significantly over the same interval (Wilcoxon signed-rank test; $p<0.001$).

Edema measurements followed a temporal pattern similar to that of the control group, with significant differences observed across all pairwise comparisons (day 0 vs. day 2, day 2 vs. day 7, and day 0 vs. day 7; all $p<0.001$).

### Between-group comparisons

The results of the between-group comparisons are summarized in Table 1.

No statistically significant difference was observed between the PBM and control groups in MMO values at any of the three time points (day 0: $p=0.195$; day 2: $p=0.251$; day 7: $p=0.860$).

Pain VAS scores on day 2 were significantly lower in the PBM group compared to the control group ($20.34\pm24.38$ vs. $41.43\pm25.15$; $p<0.001$). A statistically significant between-group difference also persisted at day 7 ($6.74\pm15.07$ vs. $9.34\pm11.16$; Mann-Whitney U; $p=0.040$); however, the absolute difference was clinically negligible (mean difference: 2.6 mm on a 100-mm VAS), as both groups had approached a near pain-free state by that time point.

Quality of life VAS scores on day 2 were significantly higher in the PBM group than in the control group ($69.83\pm25.87$ vs. $50.51\pm23.24$; $p=0.002$). By day 7, the difference between the groups was no longer significant ($p=0.154$).

Edema measurements did not differ significantly between the two groups at any time point (day 0: $p=0.791$; day 2: $p=0.473$; day 7: $p=0.897$).

## Discussion

Pain, edema, and trismus represent the most common short-term morbidities experienced by patients following the surgical extraction of impacted mandibular third molars [10, 16]. The inflammatory response and its associated complications can impair patients' daily activities and quality of life, giving rise to various social and psychological consequences [1]. In the present study, postoperative pain served as the primary outcome, on which the a-priori sample size calculation was based. Edema, trismus, and oral health-related quality of life (OHRQoL) were included as secondary outcomes; it is acknowledged that independent sample size calculations for each secondary endpoint would be required to ensure adequate statistical power for those analyses independently.

The findings of this study demonstrated that a single session of extraoral PBM applied immediately after the extraction of mandibular mesioangular impacted third molars produced a significant reduction in pain and improvement in quality of life on day 2. The pain difference persisted at a statistically significant level through day 7, albeit with negligible clinical magnitude, while the QoL advantage had resolved by that time point. No significant differences were detected for edema or trismus at any time point.

### Pain

Pain VAS scores on day 2 were significantly lower in the PBM group compared to the control group. This finding is conceptually consistent with previous studies reporting a favorable effect of PBM on early postoperative pain. In a split-mouth trial, perceived pain scores at 24, 48, and 72 h were reported to be significantly lower on the PBM-treated side [9, 17]. Conversely, another split-mouth study involving the extraction of both upper and lower third molars failed to demonstrate a statistically significant advantage in pain at days 3 and 7 [9, 17]. A multicenter, randomized, triple-blind trial reported no significant difference in any parameter, although the authors noted trends favoring PBM for pain and quality of life [10, 16].

This heterogeneous picture is also reflected in the review literature. A meta-analysis focusing on split-mouth randomized controlled trials reported a statistical trend favoring PBM for pain and swelling at 48 h and for trismus at day 7, though the certainty of evidence was rated as low [18]. Similarly, a 2021 systematic review and meta-analysis found a significant reduction in swelling but could not establish consistent significance for pain or trismus [7]. Our pain finding — a significant and clinically meaningful difference on day 2 (mean difference ~ 21 mm) that persisted at a statistically marginal but clinically negligible level on day 7 (mean

difference 2.6 mm; $p = 0.040$) — appears to be in line with reports suggesting that the analgesic benefit of PBM is most pronounced during the early postoperative period [9]. This effect is potentially mediated by the stabilization of neuronal cell membranes and the modulation of local inflammatory mediators such as COX-2 and bradykinin immediately after the surgical trauma.

## Quality of life

Quality of life VAS scores on day 2 were significantly higher in the PBM group, while this difference had resolved by day 7. The existing literature on quality of life outcomes following PBM remains inconclusive. A multicenter triple-blind trial found no statistically significant difference but acknowledged trends in favor of PBM [10, 17]. The convergence of scores by day 7 in both groups may be attributable to the natural healing process, which narrows the gap in subjective patient-reported experiences as recovery progresses.

## Edema

Edema was assessed in this study by summing the distances between three pairs of anatomical landmarks, and no significant difference between the groups was found at any time point. The flexible tape measure method and the anatomical reference points employed in this study (gonion–labial commissure, tragus–labial commissure, gonion–lateral canthus) have been used in earlier investigations [5, 13, 14].

The literature presents a notably variable picture regarding edema outcomes. On the one hand, certain studies have reported significantly lower perceived edema and facial measurements at 24, 48, and 72 h in the PBM-treated group [9, 19]. On the other hand, several split-mouth trials have observed comparable edema progression in both groups, with no statistically significant differences [17, 20]. One systematic review and meta-analysis reported significant reductions in swelling [7], while another meta-analysis demonstrated significant results in favor of PBM for pain and edema but noted that trismus remained more contentious [21]. The edema findings of the present study are consistent with those investigations that could not demonstrate a benefit of PBM on postoperative swelling [17, 20]. This discrepancy across studies is believed to arise largely from protocol variations — specifically in the number of sessions, timing of application, and route of administration [11].

## Trismus

Maximum mouth opening in both the PBM and control groups exhibited significant changes over time: a marked decrease from day 0 to day 2, followed by partial recovery toward day 7. Nevertheless, no significant difference was found between the groups at any time point. This outcome is consistent with the frequently reported inconsistency in the literature regarding trismus. While some studies have reported concurrent and significant reductions in pain, edema, and trismus [16, 22], others have failed to demonstrate a significant advantage for trismus [10, 17, 23]. A 2021 systematic meta-analysis identified a statistical benefit of PBM for pain and edema but could not establish significance for trismus [8].

## Study protocol and laser device

The GRR Laser device used in this study is a dual-wavelength system (650 + 904 nm) designed to achieve deep tissue penetration through a large-surface extraoral probe. The choice of wavelengths is critical; while 650 nm targets more superficial tissues, the 904 nm near-infrared wavelength falls within the “optical window” of biological tissue, where absorption by water and hemoglobin is minimized, allowing for deeper penetration into the mandibular bone and masseteric spaces. In a study by Baydan and Soylu (2024), the GRR Laser and the Epic10 Laser were compared following bilateral sagittal split ramus osteotomy; both devices were found to be effective in the management of lower lip paresthesia, although the GRR Laser was associated with faster recovery, which the authors attributed to its deeper penetration capacity [24]. In an animal study conducted by Karadayı et al. on rabbits, the GRR Laser and Epic10 Laser demonstrated comparable positive effects on tissue healing, with no significant difference between the two devices; the authors suggested that this finding may have been related to the limited penetration depth achievable in the rabbit model [25]. These findings informed the choice of laser device and the application procedure adopted in the present clinical study.

The PBM protocol in our study consisted of a single 10-minute extraoral session administered immediately after extraction. The current literature reveals considerable variability in application timing and number of sessions. In some studies, PBM is applied immediately after surgery and repeated on subsequent days [9, 10], while other protocols employ a single session only [20], and others still use a combined intraoral and extraoral approach [19]. A crossover study directly comparing single-session and multi-session protocols found that the two protocols yielded similar clinical outcomes, with a potential difference observed only in analgesic consumption at one specific time point [26]. Taken together, these observations underscore the extent to which protocol details can influence the outcomes of PBM studies [11].

Within the broader context of comparative therapies, the literature offers additional perspectives. In a comparison of laser therapy with triamcinolone acetonide injection, both approaches resulted in a reduction of complications from day 2 to day 7, although the magnitude of improvement was more pronounced in the triamcinolone group [27]. Findings of this nature suggest that rather than positioning PBM as a standalone superior treatment, its supportive role should be carefully weighed against the clinical context and the targeted endpoints.

## Strengths and limitations

The primary strength of this study lies in its high methodological rigor. This includes the implementation of an a-priori power analysis to ensure an adequate sample size, the use of allocation concealment, and the use of a blinded outcome assessor to minimize measurement bias. Furthermore, the strict homogeneity of the sample (asymptomatic, mesioangular, Class IB), the standardization of the surgical protocol (all operations performed by a single surgeon), and the uniform postoperative medication regimen significantly enhance the internal validity of our findings. The detailed reporting of PBM parameters (fluence, irradiance, and total energy) also ensures the reproducibility of the study protocol.

However, the study is not without certain limitations. First, the use of a single-session PBM protocol may have limited the potential therapeutic impact on facial edema and trismus, which might require cumulative dosing or multiple applications to show significant clinical differences. Second, while the outcome assessor was blinded, the lack of a sham-irradiation (placebo) group for the participants means that a potential placebo effect cannot be entirely ruled out for subjective patient-reported outcomes such as pain and quality of life. Third, although medication adherence was verified orally and via patient records, analgesic consumption was not monitored as a continuous variable. Finally, the narrow age range (17–25 years) and the focus on a single impaction type (mesioangular) strengthen the internal consistency but may limit the generalizability (external validity) of the results to older populations or more complex impaction cases.

## Conclusion

Within the constraints of a single-session extraoral protocol, dual-wavelength PBM (650 + 904 nm) applied immediately after impacted mandibular third molar extraction significantly reduced postoperative pain and improved OHRQoL on postoperative day 2, with no statistically significant effect on facial edema or trismus at any time point. A statistically significant intergroup difference in pain was also observed at day 7 ($p = 0.040$), though of negligible clinical magnitude, as both groups had reached a near pain-free state. These findings support the potential role of extraoral PBM as an adjunctive measure for early postoperative pain management; however, sham-controlled, multi-session randomized trials with longer follow-up and sufficient statistical power for secondary outcomes are needed to establish standardized clinical protocols.

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**Author contributions** A.A. and B.K.I. contributed to the study conceptualization and methodology. A.A. performed the investigation and data collection. A.A. wrote the original draft of the manuscript. B.K.I. provided supervision and validation. Z.A.B. was responsible for project administration, and involved in the review and editing of the final manuscript. All authors have read and approved the final version of the manuscript.

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**Data availability** The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethical Approval** This study was conducted in accordance with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee for Non-Drug and Non-Medical Device Research of [Anonymized University](Decision No: 2025/535, Date: January 30, 2025).

**Informed Consent** Written informed consent was obtained from all individual participants included in the study.

**Competing interest** The authors declare no competing interests.

**Clinical Trial Number** NCT07522918.

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